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AZURE VIRTUAL DESKTOP · WINDOWS 365

“It’s slow.”



Why RTT is lying to you – Reading AVD/W365 Telemetry Properly

Decomposing an AVD or Cloud PC complaint into something you can action.

Jon Hatchuel

What the ticket gives you, and what’s missing

In the ticket

- “The estimating app has been slow since Tuesday”
- One user, self-reported
- No time of day
- No idea if it is logon or in-session

What you need before you touch a console

- Getting in, or being in?
- Always, or sometimes?
- This user only, or the pool?
- Everything, or one app?

Four questions, thirty seconds, and you have eliminated roughly three quarters of the layer stack before you have opened a single blade.

“Slow” is at least eight different problems

Logon duration
Profile, GPO, scripts

In-session lag
Network or host CPU

App launch
Disk, AV, licensing

File open
SMB, sync, storage

Printing
Driver, spooler, redirect

Teams, media
Offload, codec, jitter

Screen redraw
Frames, bandwidth, GPU

Backend latency
Not the desktop at all

Eight problems. Eight different owners. Eight different fixes. One word in the ticket.

The layer map — and what natively measures each one

Layer	Who owns the fix	Native signal
Client device	User / EUC	Nothing in AVD. Intune device data for W365.
Last mile, ISP	User / network	Inferred from RTT and bandwidth estimate
Peering into Azure	Network	Inferred only — no hop visibility
Gateway region	Platform	RTT by gateway region (AVD Insights)
Transport: UDP or TCP relay	Network + platform	UdpUse column · UDP utilisation % (W365)
Host CPU, RAM, disk	Platform	Perf counters · Resource performance score
Profile layer (FSLogix)	Platform	With a custom KQL query
OS and shell startup	Platform	WVDCheckpoints stages (AVD only)
The application	App owner	Nothing / external tool
App backend, database	App owner	Nothing / external tool

Read the signature, not the symptom

THE METHOD

Pattern	Points at
Slow to get in, fine once in	Profile, GPO, logon scripts
Slow for one user every time, pool is fine	That user’s profile container
Laggy typing, RTT is good	Session host contention
Laggy typing, RTT is poor	Network, or fallback to TCP relay
Everything fine except one app	The app, or its backend
Degrades through the morning, recovers by 3pm	Density ceiling, not a fault

Every row is two metrics and a comparison. The skill is knowing which two — not knowing where the blades are.

Diagnosing Logon Fail and Poor Session Performance



`EstRoundTripTimeInMs`

`User Input Delay per Session`

`UdpUse`

`WVDConnections`

`WVDCheckpoints`

`Event log`

What the checkpoints cover, and where they stop

Measured, and reliable

- Brokering and load-balancing outcome
- RDP stack connection established
- Coarse phase boundaries to shell ready
- Total time to connect, per session

Invisible without extra work

- Which GPO is the slow one
- Which logon script is hanging
- Container attach time and its IOPS
- The client’s own pre-connection delay

You can get to “it is the profile layer” from native data. Getting to “it is this container on this storage account” means the session host event log, one host at a time.

Round-trip time measures the transport layer. Nothing above it.

A 20 ms RTT into a session host that is out of CPU produces a session that feels broken. The number will stay green the whole time the user is suffering.

`Good RTT ≠ good experience. It rules out one cause – there is considerably more to it.`

User input delay: the counter that actually tracks the complaint

```
User Input Delay per Session → instances: Max, Average, <session id>
User Input Delay per Process → instances: SessionID:PID <image.exe>
```

Why it beats CPU%

It measures how long the user's keystroke waited, not how busy the box was. A host at 60% CPU with a blocked input queue feels worse than one at 95% that is keeping up.

The 100 ms myth

100 ms is the default maximum scale on the Perfmon chart, and a configurable alert threshold in Insights. Microsoft's counter documentation publishes no official good/bad bands. Set your own and write them down.

Available on Windows 10 1809 and later, Windows 11, and Windows Server 2016 through 2025. Per-process excludes SYSTEM processes.

Transport fallback is a network problem wearing a desktop costume

UDP · Shortpath

Direct or relayed UDP. Better loss recovery, lower latency, smoother redraw under jitter.

TCP relay fallback

Works. Feels worse under loss. Silent — nobody gets an alert when a site quietly stops negotiating UDP.

How you prove it

AVD: the UdpUse column. W365: UDP utilisation % in the connection quality report. Never infer it from RTT.

Track UDP utilisation as a percentage by site, weekly. A branch that drops from 95% to 20% has had a firewall change, and you will hear about it as “the desktops got slow” three days later.

Network data logs and connection graphics data logs are preview features; enable them per host pool in diagnostic settings.

If you run both AVD and Windows 365, you’re correlating by hand

	AVD	Windows 365
Console	Azure Monitor / Insights	Intune admin center
Query language	KQL, arbitrary	Fixed reports
Logon detail	Phase checkpoints	One sign-in number
Retention	Whatever you pay for	Fixed windows, e.g. 28 days
Cost driver	Log Analytics ingestion	Included; deeper needs Intune Suite
Composite experience score	None	None

4 Clicks

The shape of data when nothing is missing

Overview

Session Host Health Status

4	0
Healthy Session hosts	Unhealthy Session hosts

Session Hosts

Total hosts	4
Idle hosts	1
PoweredOff hosts	0

2 Session Host alerts

Performance

50%

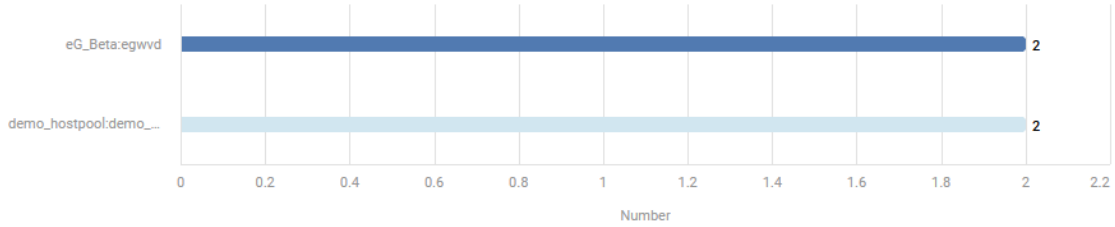
Sessions

11	1
Active	Disconnected

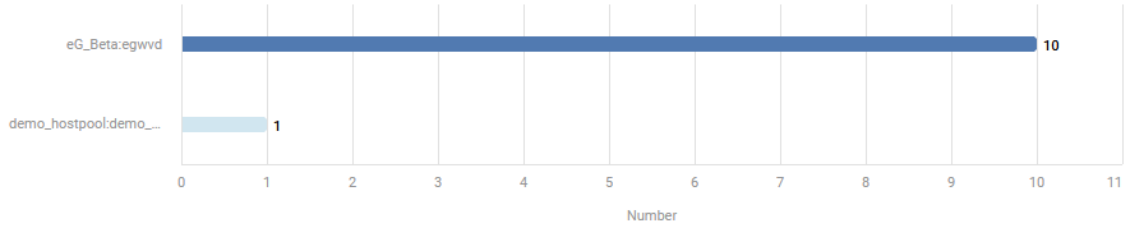
Distribution of Alerts



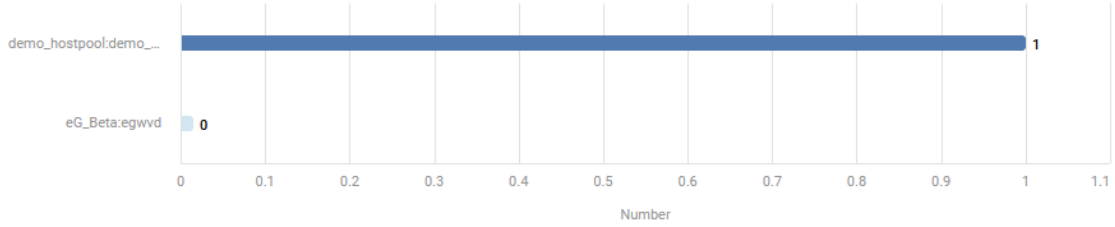
Host Pools with Most Session Hosts



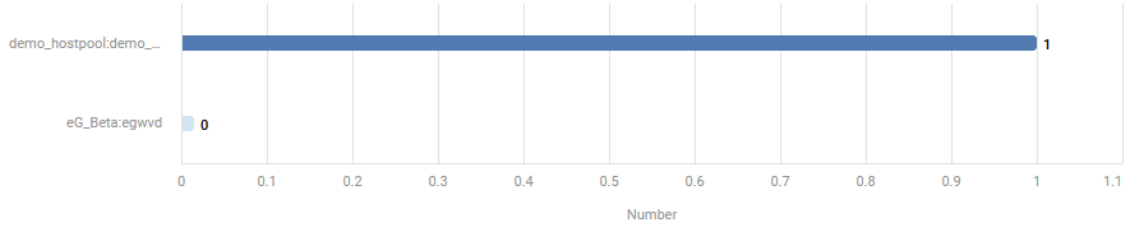
Host Pools with Most Active Sessions



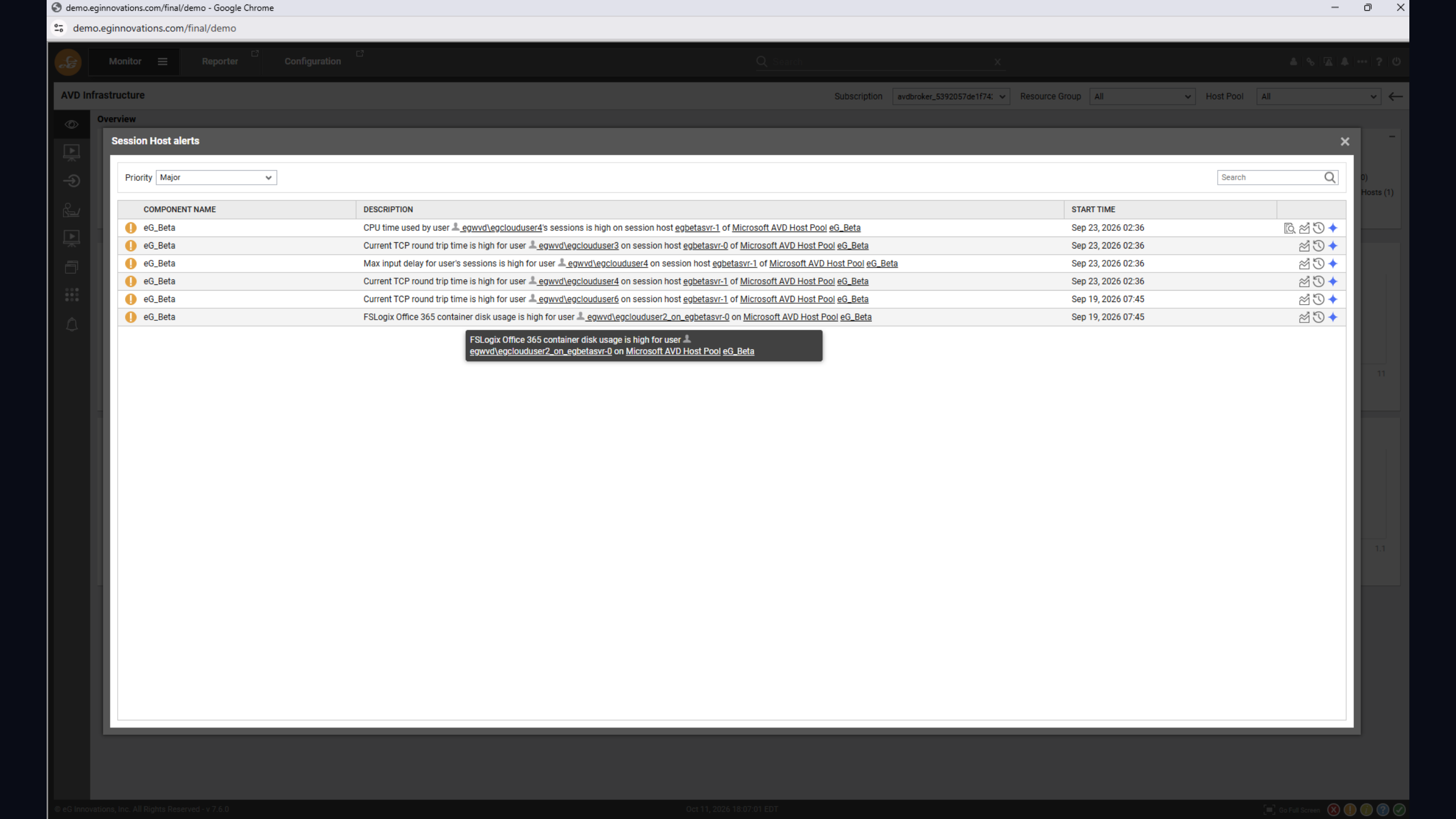
Host Pools with Most Idle Hosts



Host Pools with Most Disconnected Sessions



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demo.eginnovations.com/final/demo - Google Chrome
demo.eginnovations.com/final/demo
Monitor
Reporter
Configuration
AVD Infrastructure
Subscription
avdbroker_5392057de1f74:
Resource Group
All
Host Pool
All
Overview
Session Host alerts
Priority
Major
Search
COMPONENT NAME
DESCRIPTION
START TIME
eG_Beta
CPU time used by user egwvd\egclouduser4's sessions is high on session host egbetasvr-1 of Microsoft AVD Host Pool eG_Beta
Sep 23, 2026 02:36
eG_Beta
Current TCP round trip time is high for user egwvd\egclouduser3 on session host egbetasvr-0 of Microsoft AVD Host Pool eG_Beta
Sep 23, 2026 02:36
eG_Beta
Max input delay for user's sessions is high for user egwvd\egclouduser4 on session host egbetasvr-1 of Microsoft AVD Host Pool eG_Beta
Sep 23, 2026 02:36
eG_Beta
Current TCP round trip time is high for user egwvd\egclouduser4 on session host egbetasvr-1 of Microsoft AVD Host Pool eG_Beta
Sep 23, 2026 02:36
eG_Beta
Current TCP round trip time is high for user egwvd\egclouduser6 on session host egbetasvr-1 of Microsoft AVD Host Pool eG_Beta
Sep 19, 2026 07:45
eG_Beta
FSLogix Office 365 container disk usage is high for user egwvd\egclouduser2_on_egbetasvr-0 on Microsoft AVD Host Pool eG_Beta
Sep 19, 2026 07:45
FSLogix Office 365 container disk usage is high for user egwvd\egclouduser2_on_egbetasvr-0 on Microsoft AVD Host Pool eG_Beta

eG_Beta

Last Measurement Time : Oct 11, 2026 18:06:10

- AVD User Experience
- AVD Applications
- AVD Logon/Profile
  - FSLogix Office 365 Container Details - AVD
    - egwvd\egclouduser2_on_egbetasvr-0
    - egwvd\egclouduser1_on_egbetasvr-0
    - egwvd\egclouduser3_on_egbetasvr-0
    - egwvd\egclouduser4_on_egbetasvr-1
    - egwvd\egclouduser5_on_egbetasvr-1
    - egwvd\egclouduser6_on_egbetasvr-1
  - FSLogix Apps Event Log - AVD
    - egbetasvr-0
      - all
    - egbetasvr-1
      - all
  - FSLogix CCD Driver - AVD
    - egbetasvr-0
    - egbetasvr-1
  - FSLogix CloudCache Event Log - AVD
    - egbetasvr-0
      - all
    - egbetasvr-1
      - all
  - FSLogix Profile Details - AVD
    - egwvd\egclouduser1_on_egbetasvr-0
    - egwvd\egclouduser2_on_egbetasvr-0
    - egwvd\egclouduser3_on_egbetasvr-0
    - egwvd\egclouduser4_on_egbetasvr-1
    - egwvd\egclouduser5_on_egbetasvr-1
    - egwvd\egclouduser6_on_egbetasvr-1
  - FSLogix SMB Storage Provider - AVD
    - egbetasvr-0
    - egbetasvr-1
  - FSLogix Virtualization Driver Fileld Cache - AVD

FSLogix Office 365 Container Details - AVD -egwvd\egclouduser2_on_egbetasvr-0

FSLogix Office Container	
Is container enabled?	Yes
Container attached status	Attached success
Disk Usage	
Disk capacity (MB)	498.9805
Disk free space (MB)	10.668
Disk used space (MB)	488.3125
Disk usage (%)	97.862
Overview	
Is FSLogix application service running?	Yes
VHD access context	Attach as user
Is dynamic VHD(X)?	Yes
Is allow concurrent user sessions?	Yes
Is OneNote enabled?	Yes
Is OneNote UWP enabled?	No
Is Outlook enabled?	Yes
Is Outlook personalization enabled?	Yes
Is Teams enabled?	Yes
Is OneDrive enabled?	Yes
Is Sharepoint enabled?	Yes
Is Skype enabled?	Yes
VHD access mode	Direct access
Is Outlook cache mode enabled?	No

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eG_Beta

Last Measurement Time : Oct 11, 2026 18:04:14

- **AVD User Experience**
  - **RemoteFX User Experience - AVD**
    - egwvd\egclouduser3_on_egbetasvr-0
    - egwvd\egclouduser4_on_egbetasvr-1
    - egwvd\egclouduser6_on_egbetasvr-1
    - **egwvd\egclouduser2_on_egbetasvr-0**
    - egwvd\egclouduser5_on_egbetasvr-1
    - egwvd\egclouduser1_on_egbetasvr-0
  - Users - AVD
    - egwvd\egclouduser4_on_egbetasvr-1
    - egwvd\egclouduser1_on_egbetasvr-0
    - egwvd\egclouduser2_on_egbetasvr-0
    - egwvd\egclouduser3_on_egbetasvr-0
    - egwvd\egclouduser5_on_egbetasvr-1
    - egwvd\egclouduser6_on_egbetasvr-1
  - Network Performance - AVD
    - egwvd\egclouduser1_on_egbetasvr-0
    - egwvd\egclouduser2_on_egbetasvr-0
    - egwvd\egclouduser3_on_egbetasvr-0
    - egwvd\egclouduser4_on_egbetasvr-1
    - egwvd\egclouduser5_on_egbetasvr-1
    - egwvd\egclouduser6_on_egbetasvr-1
- AVD Applications
- AVD Logon/Profile
- AVD Sessions
- AVD Session Host
- AVD Processes/Services
- AVD Network/TCP
- AVD Operating System
- AVD Storage

RemoteFX User Experience - AVD -egwvd\egclouduser2_on_egbetasvr-0

Sessions	
User sessions (Number)	1
User's connection quality indicator	Weak connection
Graphics	
Average frames encoding time (Milliseconds)	4
Frame quality (%)	100
Frames skipped due to insufficient client resources (Frames/sec)	0
Frames skipped due to insufficient network resources (Frames/sec)	0
Frames skipped due to insufficient server resources (Frames/sec)	0
Graphics compression ratio (%)	1
Input frames (Frames/sec)	0
Output frames (Frames/sec)	0
Source frames (Frames/sec)	0
Network	
Forward error correction rate (%)	0
Loss (%)	0
Retransmissions (%)	0
Bandwidth	
Current TCP bandwidth (Mbps)	7.731
Current UDP bandwidth (Mbps)	0
TCP	
Base TCP round trip time (Seconds)	0.22
Current TCP round trip time (Seconds)	0.229
TCP received rate (Kbps)	0
TCP sent rate (Kbps)	5.535
UDP	
Base UDP round trip time (Seconds)	0
Current UDP round trip time (Seconds)	0
UDP received rate (Kbps)	0
UDP sent rate (Kbps)	0

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The diagnostic runbook

TAKE THIS WITH YOU

Complaint	First metric	What it rules in
Slow to get in	Checkpoint phase split · sign-in time	Profile, GPO, scripts, VM start
Laggy in session	Input delay per session, Max instance	Host contention vs network
Laggy, input delay is fine	RTT P90 · UdpUse · bandwidth estimate	Last mile, or transport fallback
One app only	Input delay per process, by image	That app, or its backend
Whole pool, mornings	Resource score · CPU/RAM spike time	Density ceiling, sizing decision
One site, since Tuesday	UDP utilisation % trend, by site	Somebody changed a firewall
Nothing correlates	Client device · app backend	The two layers you cannot see